

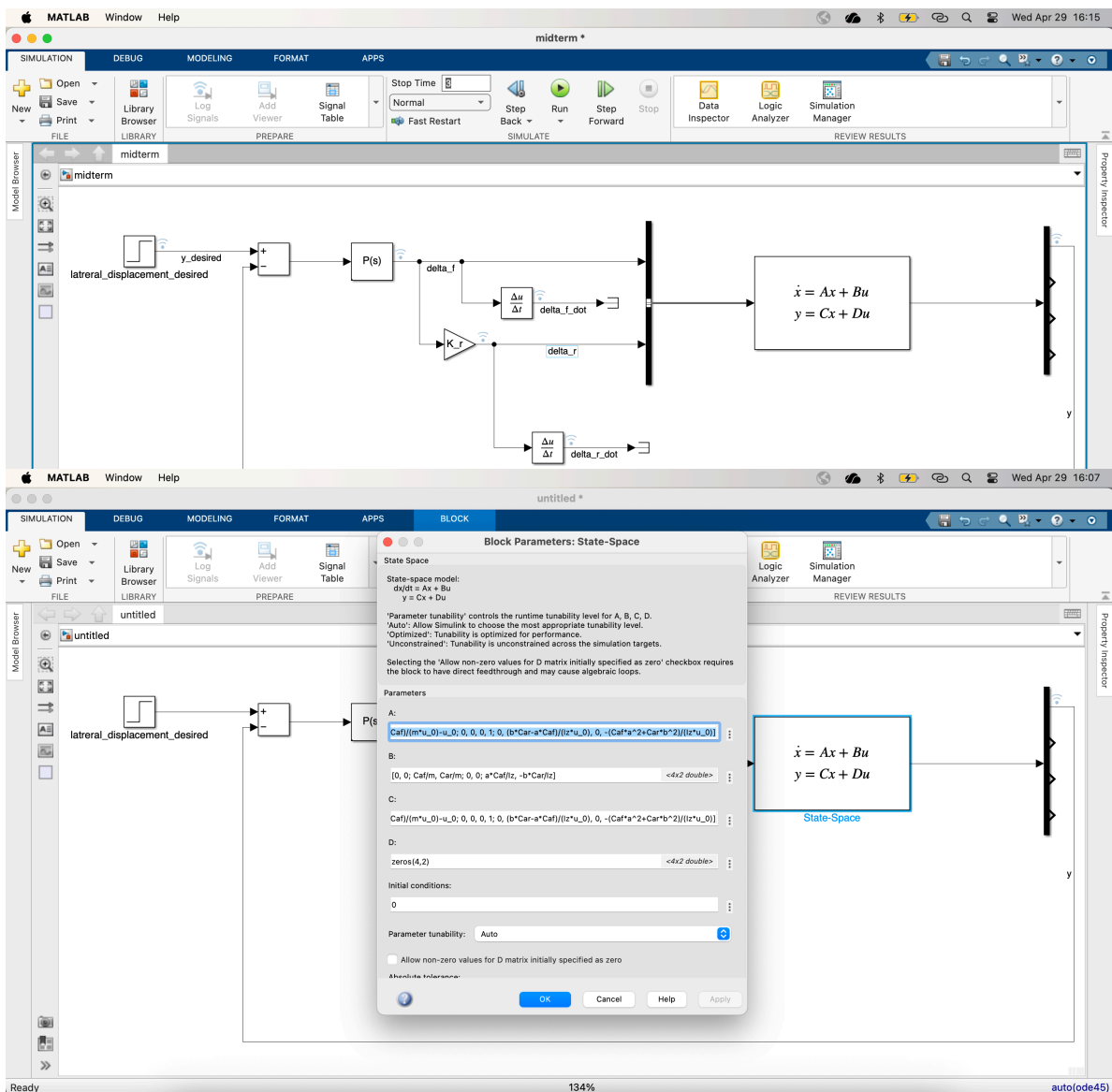
MMU753
MIDTERM
SIMULINK PART
YUSUF AĞRI ÖĞÜT
N25240126

Model

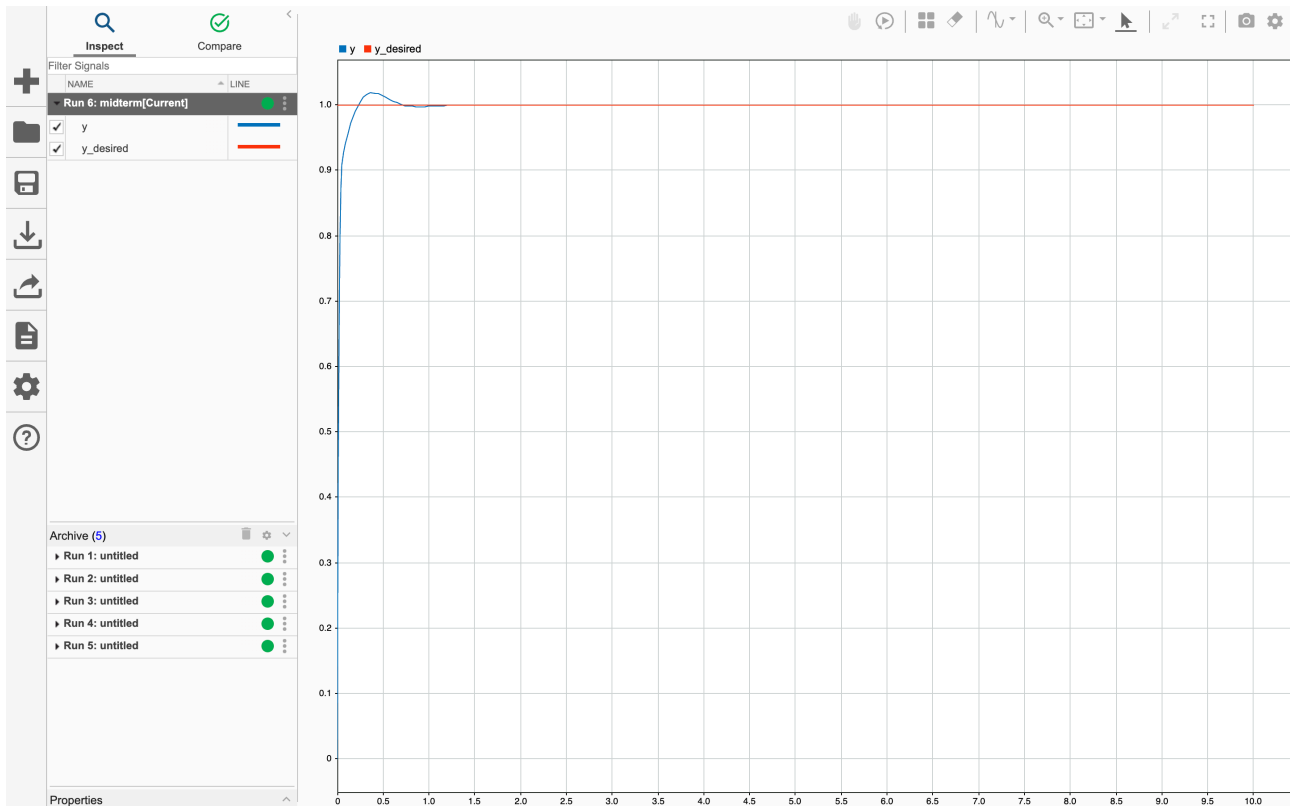
First, parameters given in question are written into a matlab script called 'init.m'. Constant velocity is chosen as 15 m/s. Calculated K_r value is 0.2171.

```
Editor - /Users/yusufcagriogut/Documents/GitHub/mmu753/midterm/init.m
init.m x +
1  clf; clear all; clc;
2  %%
3  a=1.14;
4  L=2.54;
5  m=1500;
6  Iz=2420;
7  Caf=44000*2;
8  Car=47000*2;
9  b=L-a;
10 g= 9.81;
11 u_0 = 15;
12
13 K_r = (m*u_0^2*a-b*(a+b)*Car)/...
14       (m*u_0^2*b-a*(a+b)*Caf)*...
15       Caf/Car;
```

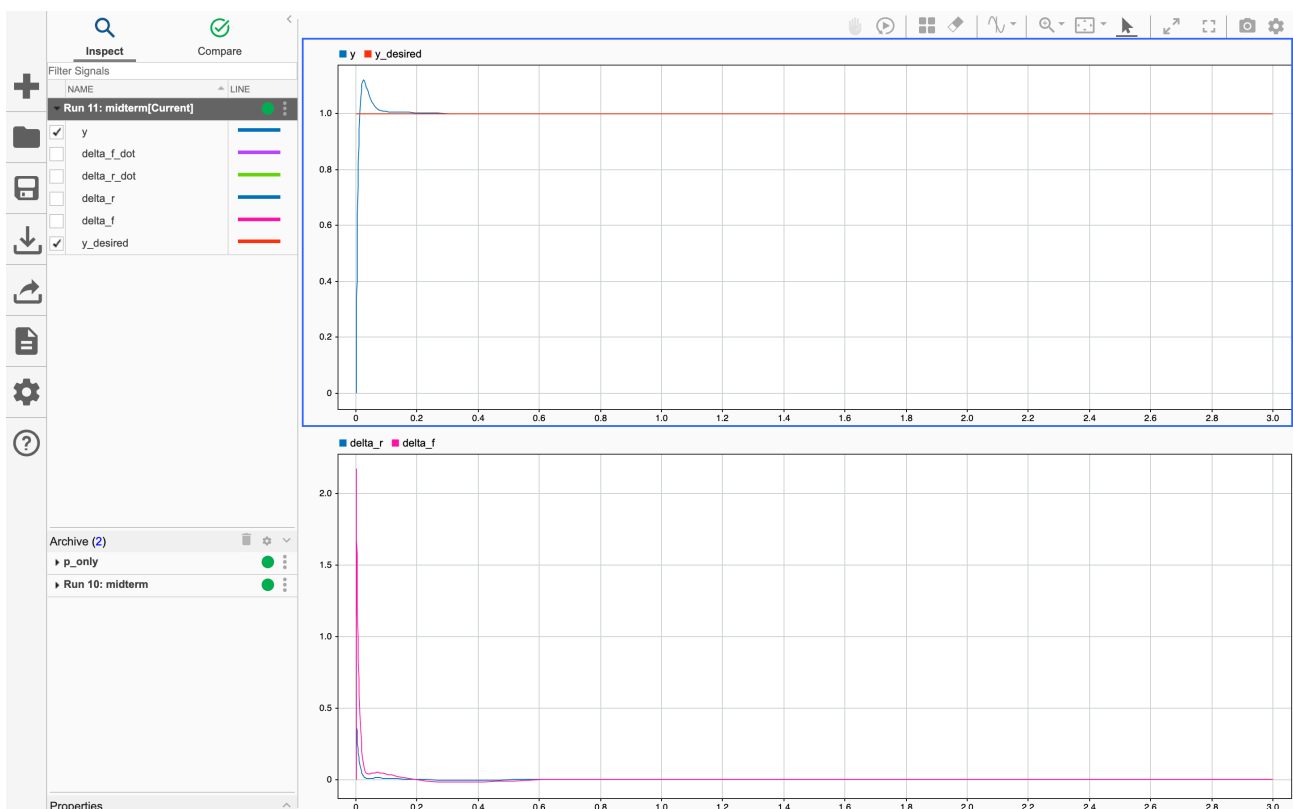
Following, given state-space model is implemented in Simulink.

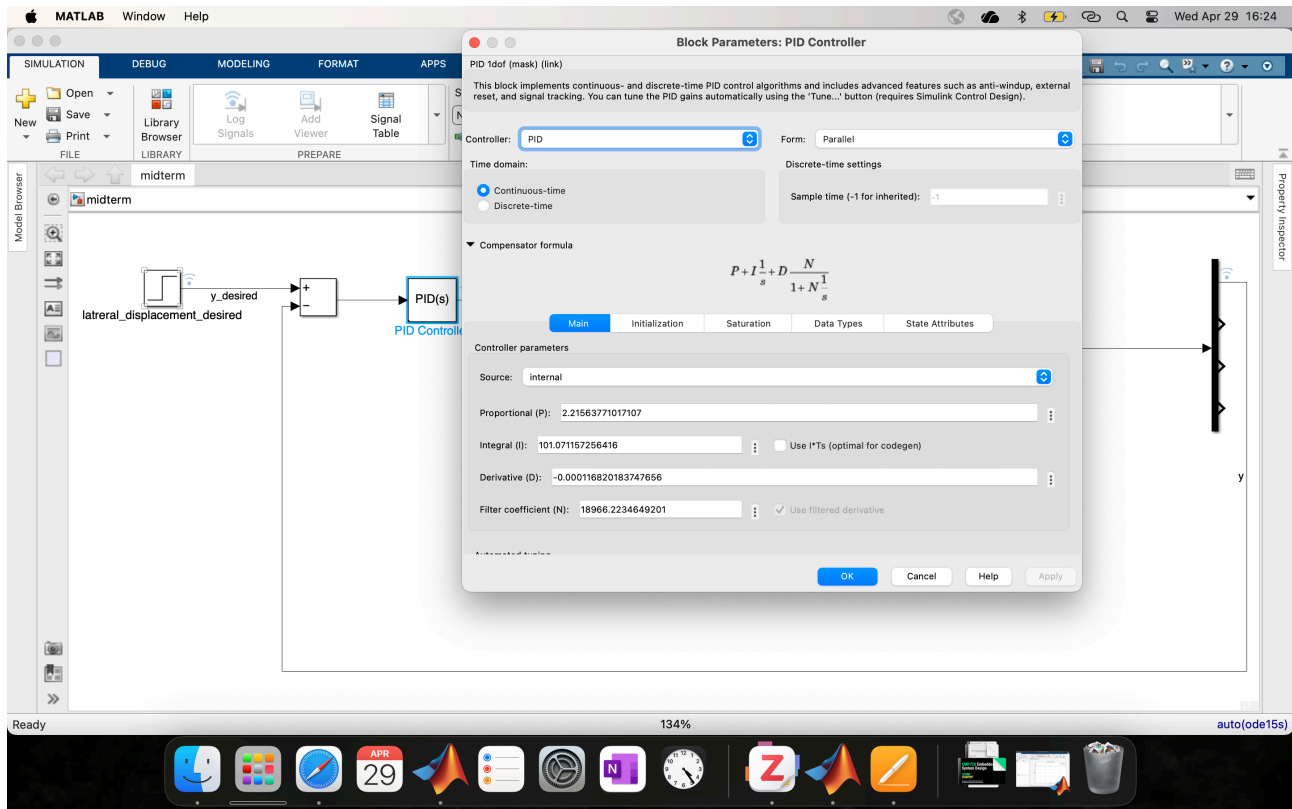


Then, in order to check if model and simulation is working properly, a very basic only P controller is added to see result. Below is the resulting lateral displacement plot. It can be seen that model works fine.

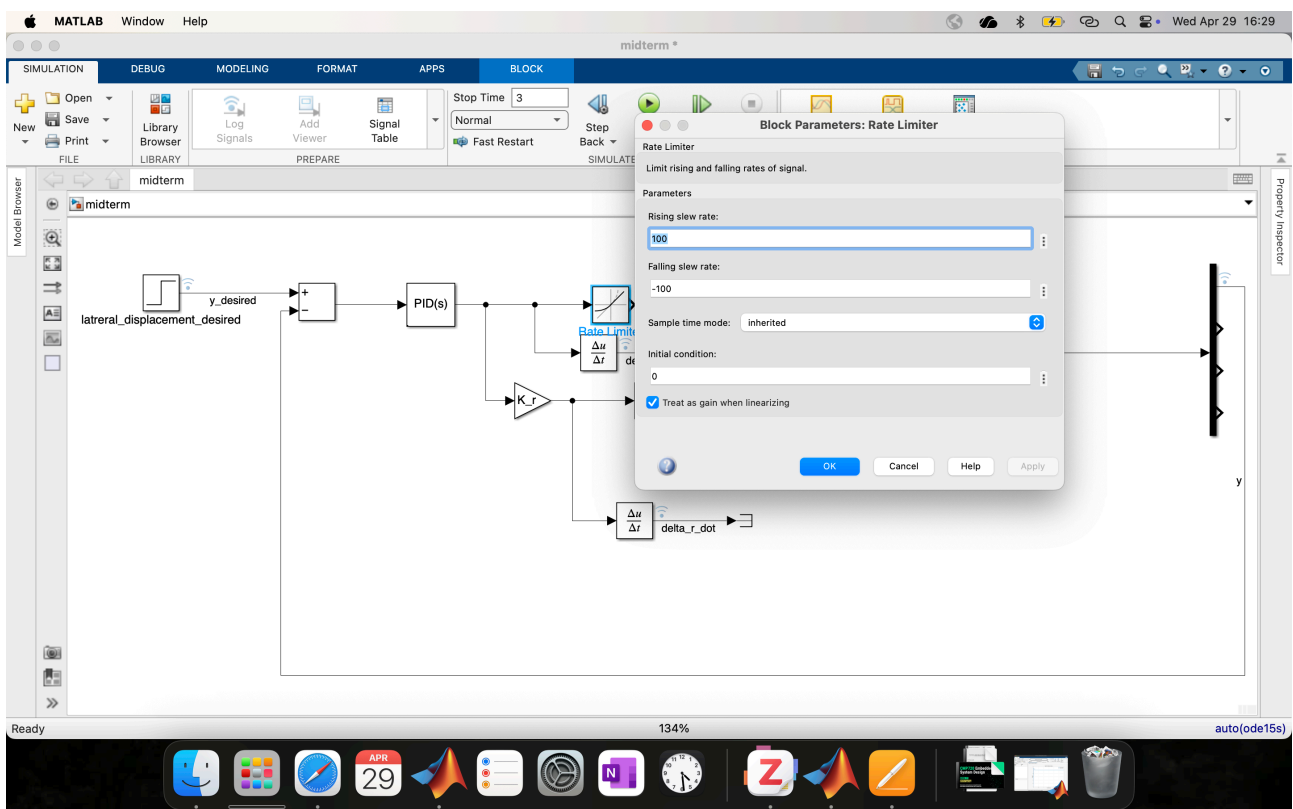
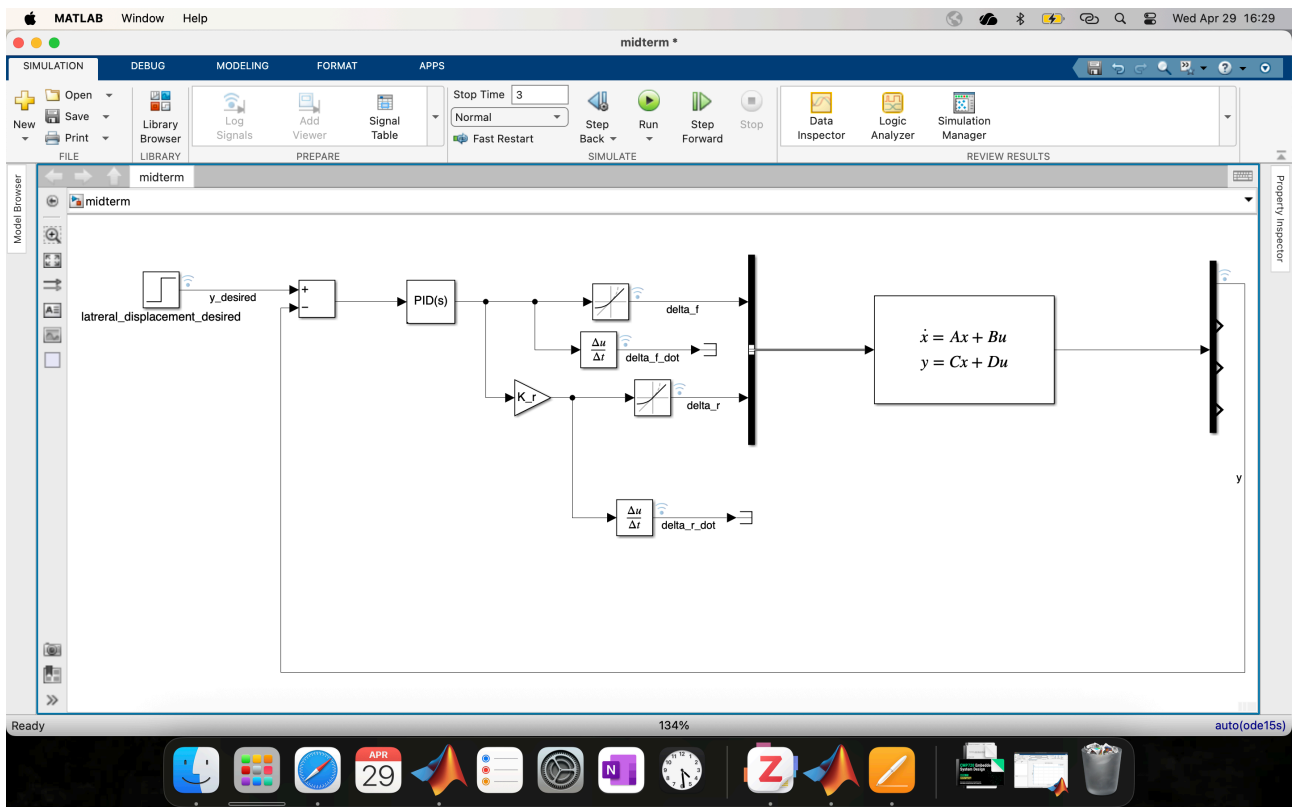


Then, a more complex controller is tried. PID controller is selected and it is tuned using PID tuner.

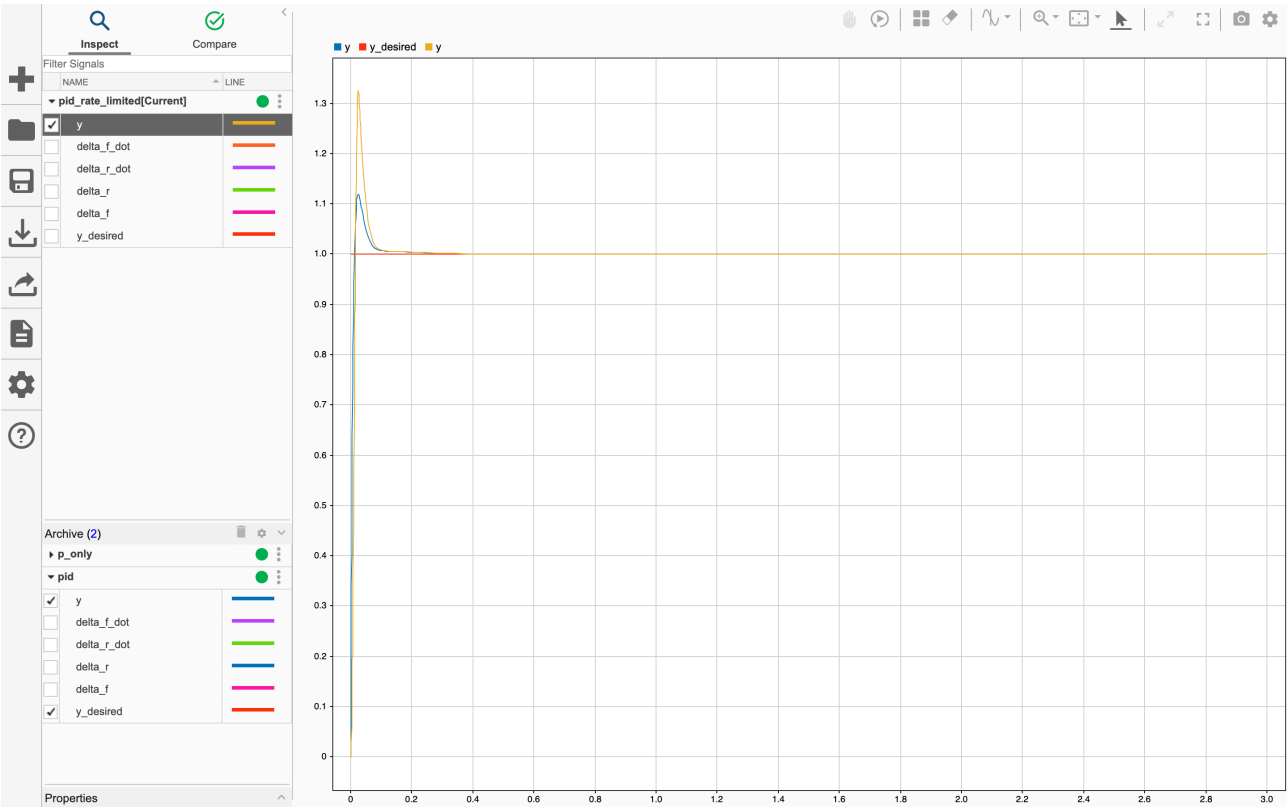




Even though result looks good, steering angle rates are too high and this is not realistic. Steering angle rates are limited to 100 and -100 degrees per second in model, using rate limiter.



In following graph, before and after rate limiter system response can be seen (yellow is after rate limit):



It can be seen that realistic wheel and steering dynamics is very important and tuning should be done considering that.